Melodius

Twin Piano Experience

Connecting through Music

Group 6-GPP

What is Melodius?

A revolutionary way to connect through synchronized music anytime, anywhere.

MelodiUS connects people through sound.
Music is a powerful way to express emotion without relying on words. Through music, MelodiUS brings people together across distance, time, and place.
It creates shared musical experiences by letting people communicate and bond through melodies made for one another.

Why a Piano?

The piano is a widely loved instrument. However, meaningful musical experiences have traditionally required people to be physically together.

Melodius reimagines the piano as a collaborative instrument. It enables people to play, share and create music together, whether they are nearby or far apart.

This opens the door to shared melodies, remote jam sessions and bonding without distance restriction.

Primary Target Users?

A shared musical experience across miles, connecting people.

User Personas:

Beloved ones

- Friends
- Families
- Couples

Research

Melodius builds on research in social bonding, embodied interaction, and IoT system design by combining synchronous music-making, to create a shared sense of presence across distance.

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1. Most current remote music tools (e.g. video calls, collaborative DAWs) rely on screen-based, cognitively demanding interaction. Melodius improves on this by offering co-temporal, embodied sound-making, which fosters emotional closeness without requiring language or attention-heavy interfaces (Tarr, B., Launay, J. and Dunbar, R. (2014).
2. Existing connected devices often treat each object as autonomous, leading to turn-taking rather than true collaboration. Melodius improves on this by maintaining a shared musical state (timing, rhythm, scale), enabling both users to act within the same sonic space (Dourish, P. (2001).
3. Streaming audio across long distances is highly sensitive to network latency and packet loss, which degrades musical experience. Twin Piano addresses this limitation by transmitting lightweight event data (key pressed, timing, velocity), which significantly reduces bandwidth and improves robustness (Al-Fuqaha, A. et al. (2015).
4. How Sound Connects Us - https://mnartists.walkerart.org/resonating-relating-how-sound-connects-us Emily Dzieweczynski, Resonating/Relating:
5. How Humans Communicate Through Sound : Kraus, Nina, Slater, Jessica, Beyond Words:

How It Works

KEY PRESS

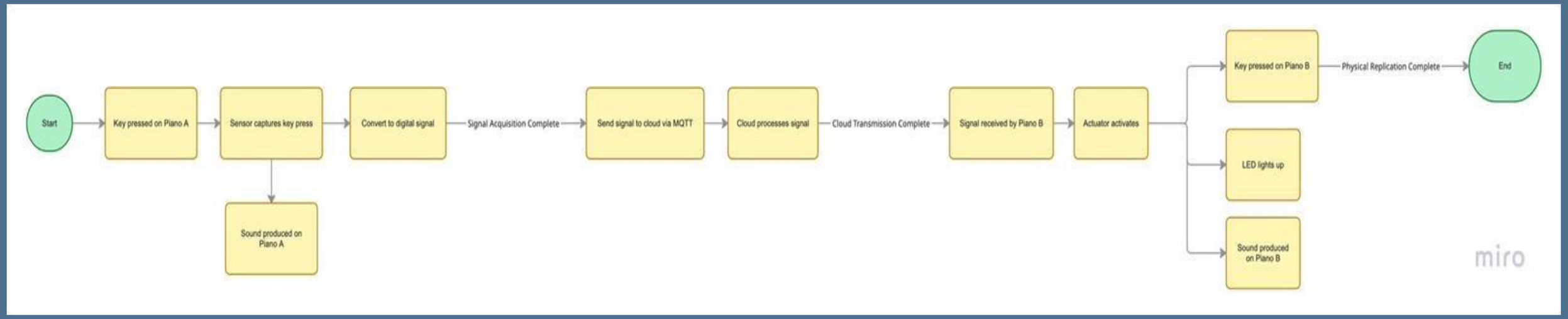
When a user presses a key on Piano A, the sensor immediately detects the action and prepares for signal transmission.

SIGNAL DETECTION

The sensor sends a signal via the IoT module, ensuring real-time communication and connection with the counterpart piano across distances.

INSTANT PLAYBACK

The signal reaches Piano B, lighting up the corresponding key, activating movement, and producing sound simultaneously, creating a synchronized experience.



Technology Components Hardware

The Melodius piano utilizes **microcontrollers, sensors,** and LEDs to ensure precise key detection and synchronized lighting effects during play.

Mini Keys

Sensitive keys for interactive musical input and expressive play

Led Lighting

Integrated LEDs provide real-time visual feedback for key presses and synchronization

Microcontrollers with Wifi

ESP32 (or similar) enables key detection, processing, and wireless communication between devices.

Sensors

RFI-resistant touch sensor detect key presses and interaction intensity

Servos

Used to mirror physical key movement on the paired piano for enhanced synchronisation

Technology Components
Software

Real-time messaging protocols manage data transmission between pianos, enabling seamless synchronization despite varying internet conditions and minimizing latency challenges.

Microcontroller Code

Handles key detection, LED control, and communication logic on each device

Wireless Communication Protocol

Wifi or MQTT to enable low-latency transmission of key events between pianos.

Synchronization Logic

Ensures key presses, timing, and lighting remain aligned across both devices

Sound Generation Module

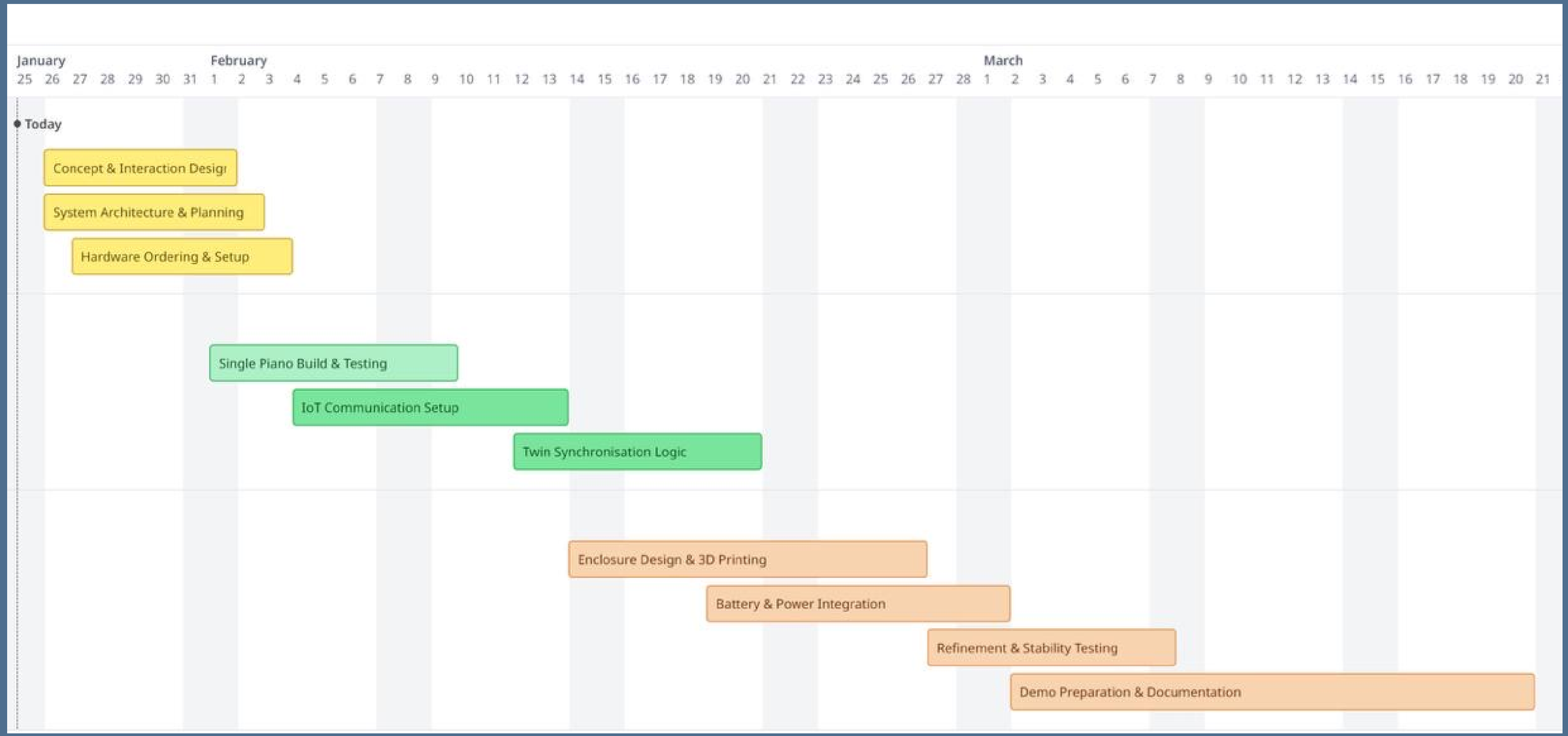
Maps key inputs to musical notes and produces local audio playback

Power & State Management

Controls sleep modes, connectivity states, and efficient battery usage

High Level Project Timeline

The project aim to follow an Eight-week delivery plan



Next steps and current challenges

Current Challenges

- Network latency may affect real-time key synchronisation.
- Mechanical or sensor-based key detection requires careful calibration.
- Battery life is limited when using Wi-Fi, LEDs, and sound together.
- Balancing simplicity for non-musicians with expressive musical interaction.
- Ensuring reliable touch input without false triggers from electrical noise.
- Physical enclosure constraints limit component placement and heat dissipation.

Next Steps

- Optimise data transmission using lightweight, event-based messaging.
- Refine key sensitivity and LED feedback for clearer interaction.
- Improve power management through sleep modes and efficient components.
- Conduct small-scale user testing to evaluate emotional engagement.
- Iterate enclosure design for ergonomics and durability.
- Documentation